Emergent Gaussian channels for Pauli noise: certified quantum-capacity thresholds and positive capacity of the additive Gaussian noise channel beyond $1/\sqrt{e}$

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A qubit Pauli channel acting on permutation-symmetric states of M qubits with a bounded number of excitations around a pole of the Bloch sphere behaves, for large M, like a single-mode bosonic Gaussian channel; Ahmed *et al.* identified this emergent channel for the depolarizing channel. We give its form for an arbitrary Pauli channel with Bloch factors $(\lambda_a, \lambda_b, \lambda_c)$ and pole along c: $X = \mathrm{diag}(\lambda_a, \lambda_b)/\sqrt{\lambda_c}$ and $Y = \mathrm{diag}(1-\lambda_a^2, 1-\lambda_b^2)/(2\lambda_c)$. Its complete positivity is equivalent to the nonnegativity of the Gram determinant $1-\lambda_a^2-\lambda_b^2-\lambda_c^2+2\lambda_a\lambda_b\lambda_c$. For the independent X–Z (BB84) channel the emergent channel is the additive Gaussian noise (AGN) channel; for the 2-Pauli channel it is a quantum-limited amplifier after pure loss. The emergent channel is antidegradable if and only if $\lambda_\perp \le 1/\sqrt{2}$, the shrinking factor of optimal phase-covariant cloning, which caps what this route can certify. Using an exact band-decomposed evolution in the Dicke basis and two interval-arithmetic certifiers, we certify positive coherent information of explicit M-qubit states at $q = 0.126$ for BB84 ($M = 240$) and $p = 0.1255$ for the 2-Pauli channel ($M = 240$), above the previously reported 0.118371 and 0.118067, and at total error 0.1977936 for the depolarizing channel ($M = 240$). In the Gaussian limit we certify the emergent depolarizing channel at total error 0.199546875; with the convergence theorem of Ahmed *et al.* this is, to our knowledge, the largest depolarizing threshold bound whose final sign is certified (the larger 0.2029161 rests on a floating-point value $I_c \approx 10^{-12}$). Transferring the BB84 code to the Gaussian limit, we certify positive coherent information for the AGN channel at noise variance $\sigma^2 = 163/400$ ($\sigma \approx 0.6384$), hence positive quantum capacity for all $\sigma^2 \le 0.4075$, beyond $\sigma^2 = 1/e$ where the Gaussian-input coherent information vanishes; this refutes the 2001 speculation that the Gaussian-input coherent information is the capacity of this channel. For the thermal attenuator we certify $Q > 0$ well inside the open window at every temperature tried (e.g. $\eta \ge 0.78$ with one thermal photon, below the recently certified 0.7841). All results are computer-assisted; human verification is pending.

I. INTRODUCTION

The quantum capacity of a channel $\mathcal{N}$ is its regularized coherent information [1, 2],

$$Q(\mathcal{N}) = \lim_{n\to\infty} \frac{1}{n} \max_{\rho} I_c(\rho, \mathcal{N}^{\otimes n}), \tag{1}$$

with $I_c(\rho,\mathcal{N}) = S(B) - S(RB)$, where $S(B) = S(\mathcal{N}(\rho))$, $S(RB) = S((\mathrm{id}_R \otimes \mathcal{N})(\psi_\rho))$ and ψ_ρ purifies ρ. Any explicit input ρ on M channel uses with $I_c(\rho, \mathcal{N}^{\otimes M}) > 0$ certifies $Q(\mathcal{N}) \ge I_c/M > 0$. For a one-parameter noise family $\mathcal{N}_p$ we write $p_{\mathrm{th}} = \sup\{p : Q(\mathcal{N}_p) > 0\}$. Because the maximal coherent information can be strictly superadditive [3], p_{th} can lie above the hashing point [4], where the single-letter coherent information of the maximally mixed input vanishes; the improvements come from degenerate codes [3, 5–7].

We consider qubit Pauli channels $\mathcal{N}(\rho) = p_I\rho + p_X X\rho X + p_Y Y\rho Y + p_Z Z\rho Z$ and three standard families: the depolarizing channel $(p_I, p_X, p_Y, p_Z) = (1-3p, p, p, p)$, whose total error is $3p$; the independent X–Z channel, also called the BB84 channel, $((1-q)^2, q(1-q), q(1-q), q^2)$; and the 2-Pauli channel $(1-2p, p, 0, p)$. Their hashing points are $3p = 0.189290$, $q = 0.110028$ and $p = 0.113546$.

State of the art. Bhalerao and Leditzky [8] introduced representation-theoretic methods for permutation-invariant codes. Agarwal *et al.* [9] extended them to rank-two states on the full symmetric subspace and reported $q = 0.118371$ (BB84) and $p = 0.118067$ (2-Pauli) at input length $n = 30$, and $p = 0.064657$ per Pauli for the depolarizing channel at $n = 45$. Krohn-Grimberghe [10] gave exact-arithmetic, independently machine-checkable positivity proofs for the depolarizing channel, at $p = 16239/250000$ per Pauli ($3p = 0.194868$, $n = 45$), and noted that the final sign evaluation of earlier reported thresholds was done in floating point. Ahmed *et al.* [11] showed that the depolarizing channel, acting on symmetric states with a bounded number of excitations, converges to a bosonic thermal attenuator; optimizing residue (cat-like) codes for that Gaussian channel and using a convergence theorem (their Theorem 6), they obtained $p_{\mathrm{th}} \gtrsim 0.2029161$ in total error (transmissivity $\eta = 0.7294452$), valid for sufficiently large M, with a reported Gaussian-channel coherent information $I_c \approx 10^{-12}$ at that point. Table I summarizes.

The additive Gaussian noise channel. The AGN channel $\mathcal{N}_\sigma$ (also called the Gaussian random displacement or classical-noise channel) displaces both quadratures by independent Gaussian noise of variance σ^2, in units where the vacuum variance is 1/2. Its quantum capacity is unknown. Thermal (Gaussian) inputs give $Q(\mathcal{N}_\sigma) \ge \max\{0, \log_2[1/(e\sigma^2)]\}$ [12], a rate that Gottesman–Kitaev–Preskill (GKP) lattice codes can match [13]; the channel is antidegradable, hence $Q = 0$, for $\sigma^2 \ge 1/2$ [14, 15]. Lin and Noh [16] decoded multi-

TABLE I. Lower bounds on p_{th}. "This work" entries are certified for an explicit state (Sec. VI); the last column is the ceiling of the Gaussian-limit route (Sec. III). For the AGN channel the parameter is σ^2 and "hashing" is the Gaussian-input value.

channel	hashing	previous best	this work	ceiling
BB84 (q)	0.110028	0.118371 [9]	0.126	0.146447
2-Pauli (p)	0.113546	0.118067 [9]	0.1255	0.146447
depol. ($3p$)	0.189290	0.2029161[a] [11]	0.1995469[b]	0.219670
AGN (σ^2)	$1/e$	$1/e$ [12]	0.4075	1/2

[a] Gaussian limit with a convergence theorem, M sufficiently large; the best previous certified value for an explicit size is 0.194868 ($n = 45$) [10].

[b] Gaussian limit with I_c certified (Sec. VI B) and the convergence theorem of Ref. [11], M sufficiently large; the best explicit-size certificate is 0.1977936 ($M = 240$).

mode GKP codes with exact or approximate maximum-likelihood decoders: the $d = 39$ surface-square code keeps a nonzero rate up to $\sigma = 0.598$, the threshold of the surface-square family is close to $1/\sqrt{e} \simeq 0.6065$, and they discuss the prospect of a nonzero rate in the window $1/\sqrt{e} \le \sigma < 1/\sqrt{2}$, where no positive lower bound was known. Zhang, Wu and Guo [17] reported a code-capacity threshold $\sigma \approx 0.67$ for rectangular GKP codes concatenated with the XZZX surface code, from simulations with minimum-weight matching up to distance 9; Lin and Noh found that the crossing points drift downward with the distance and concluded that this does not establish a threshold above $1/\sqrt{e}$ [16]. Such simulations are not proofs. Harrington and Preskill, observing that GKP rates match the Gaussian-input coherent information, wrote that "it is tempting to speculate that this optimal one-shot coherent information actually is the quantum capacity of the channel" [13]. The closest related work is by Mele *et al.* [18], who prove that non-Gaussian inputs beat the Holevo–Werner thermal-state bound for the thermal attenuator.

Contributions. (i) We give the emergent Gaussian channel of an arbitrary Pauli channel (Sec. II), show that its complete positivity is the nonnegativity of a Gram determinant, and identify the BB84 and 2-Pauli cases. (ii) We show that the emergent channel is antidegradable exactly when $\lambda_\perp \le 1/\sqrt{2}$ (Sec. III). (iii) We describe an exact finite-M method (Sec. IV) and two rigorous certifiers based on ball arithmetic (Sec. V). (iv) We certify thresholds for the BB84, 2-Pauli and depolarizing channels and for a family between the last two (Sec. VI). (v) We certify $Q(\mathcal{N}_\sigma) > 0$ for $\sigma^2 \le 0.4075$ (Sec. VII). Section VIII lists what did not work, Sec. IX open problems and Sec. X how to reproduce every certificate.

Status. The emergent-channel formula is derived from first and second moments; its Gaussianity is proven only for the depolarizing channel [11] and is supported here numerically. The thresholds of Sec. VI and the AGN result use no limit theorem: each is a finite computation with rigorous error control. They were checked by two independent code reviews and, for the AGN result, by an independent re-implementation, all carried out by separate instances of an AI system (see Acknowledgments). Independent human verification is pending.

II. EMERGENT GAUSSIAN CHANNEL OF A PAULI CHANNEL

A. Setting

A Pauli channel shrinks the Bloch vector along the coordinate axes by the factors

$$\lambda_x = 1 - 2(p_Y + p_Z) \quad \text{and cyclic permutations.} \tag{2}$$

Let (a, b, c) be a permutation of (x, y, z) with $\lambda_c > 0$; the axis c will be the pole. Let $J_a = \frac{1}{2}\sum_i \sigma_a^{(i)}$ be the collective spin of M qubits and $|J, m\rangle$ the Dicke states with m measured along c. Symmetric inputs live in the $J = M/2$ block, and $|M/2, M/2 - k\rangle$ has k excitations below the pole. For $k \ll M$ the Holstein–Primakoff map [19] $J_- \simeq \sqrt{2J}\, b$ identifies $|M/2, M/2 - k\rangle$ with the Fock state $|k\rangle$ of a mode b, with quadratures $x = (b + b^\dagger)/\sqrt{2}$, $p = i(b^\dagger - b)/\sqrt{2}$ and vacuum variance $1/2$.

The output $\mathcal{N}^{\otimes M}(\rho)$ of a symmetric input is permutation invariant, so by Schur–Weyl duality

$$\begin{aligned} \mathcal{N}^{\otimes M}(\rho) &= \bigoplus_J B_J \otimes \frac{\mathbb{1}_{d_J}}{d_J}, \\ (\mathrm{id}_R \otimes \mathcal{N}^{\otimes M})(\psi_\rho) &= \bigoplus_J (RB)_J \otimes \frac{\mathbb{1}_{d_J}}{d_J}, \end{aligned} \tag{3}$$

where d_J is the multiplicity of spin J and B_J carries the full weight of sector J. The multiplicity terms $\mathrm{Tr}(B_J)\log d_J$ appear identically in $S(B)$ and $S(RB)$, so

$$I_c(\rho, \mathcal{N}^{\otimes M}) = \sum_J \big[S(B_J) - S((RB)_J)\big]. \tag{4}$$

For inputs with few excitations the output weight concentrates at $J \simeq \lambda_c M/2$. In each sector we use the Holstein–Primakoff mode around the same pole.

B. Moment formula

We use the conventions of Ref. [20]: a single-mode Gaussian channel acts on the mean vector d and covariance matrix V as $d \mapsto Xd$, $V \mapsto XVX^{\mathsf{T}} + Y$, and it is completely positive if and only if

$$Y + \tfrac{i}{2}\,(\Omega - X\Omega X^{\mathsf{T}}) \succeq 0, \qquad \Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{5}$$

Pure loss $\mathcal{L}_T$ has $X = \sqrt{T}\mathbb{1}$, $Y = \frac{1-T}{2}\mathbb{1}$; the quantum-limited amplifier $\mathcal{A}_G$ has $X = \sqrt{G}\mathbb{1}$, $Y = \frac{G-1}{2}\mathbb{1}$; the AGN channel $\mathcal{N}_\sigma$ has $X = \mathbb{1}$, $Y = \sigma^2\mathbb{1}$.

Proposition 1 (Emergent channel; moment formula)**.** *For inputs supported on a fixed number of excitations around the pole c, the first and second moments of the renormalized output mode, sector by sector, converge as $M \to \infty$ to those of the Gaussian channel*

$$X = \frac{1}{\sqrt{\lambda_c}}\,\mathrm{diag}(\lambda_a, \lambda_b), \qquad Y = \frac{1}{2\lambda_c}\,\mathrm{diag}(1-\lambda_a^2,\, 1-\lambda_b^2). \tag{6}$$

Derivation. (i) First moments. In the Heisenberg picture $\mathcal{N}^\dagger(\sigma_a) = \lambda_a \sigma_a$, so $\langle J_a\rangle \mapsto \lambda_a \langle J_a\rangle$. With $J_a \simeq \sqrt{J}\,x$ in a sector of spin J, $J_{\mathrm{in}} = M/2$ and $J_{\mathrm{out}} \simeq \lambda_c M/2$, this gives $x \mapsto (\lambda_a/\sqrt{\lambda_c})\,x$ and likewise $p \mapsto (\lambda_b/\sqrt{\lambda_c})\,p$. (ii) Second moments. The vacuum (all spins at the pole) is mapped to the product state $\left[\frac{1}{2}(\mathbb{1} + \lambda_c \sigma_c)\right]^{\otimes M}$. Inside each spin-$J$ sector this state is proportional to r^k, with $k = J - m$ and $r = (1-\lambda_c)/(1+\lambda_c)$. This is a truncated geometric distribution with mean $r/(1-r) = (1-\lambda_c)/(2\lambda_c)$, that is, a thermal state with isotropic quadrature variance $1/(2\lambda_c)$. Then $Y = V_{\mathrm{out}}(\mathrm{vac}) - \frac{1}{2} X X^{\mathsf{T}}$ gives Eq. (6). □

The formula fixes the channel once the limit is known to be Gaussian. Ahmed *et al.* proved this for the depolarizing channel [11]. For other Pauli channels we have no proof; the exact computations of Table II support it (Sec. II D).

Lemma 1 (Complete positivity)**.** *For the pair (6), condition (5) is equivalent to*

$$\Delta := 1 - \lambda_a^2 - \lambda_b^2 - \lambda_c^2 + 2\lambda_a\lambda_b\lambda_c \;\geq 0, \tag{7}$$

which holds for every Pauli channel.

Proof. For 2×2 matrices $\Omega - X\Omega X^{\mathsf{T}} = (1 - \det X)\Omega$, so Eq. (5) reads $Y \succeq 0$ and $\det Y \geq \frac{1}{4}(1-\det X)^2$. With Eq. (6) the latter is $(1-\lambda_a^2)(1-\lambda_b^2) \geq (\lambda_c - \lambda_a\lambda_b)^2$, which is $\Delta \geq 0$; $Y \succeq 0$ holds since $|\lambda_a|, |\lambda_b| \leq 1$. Δ is the determinant of the correlation (Gram) matrix $C = \left(\begin{smallmatrix} 1 & \lambda_c & \lambda_b \\ \lambda_c & 1 & \lambda_a \\ \lambda_b & \lambda_a & 1 \end{smallmatrix}\right)$. C is affine in the Pauli probabilities, and at the four Pauli channels $\rho \mapsto P\rho P$ it equals vv^{T} with $v \in \{\pm 1\}^3$. For example, $P = X$ gives $(\lambda_x, \lambda_y, \lambda_z) = (1, -1, -1)$ and $v = (1, -1, -1)$. Hence $C = \sum_P p_P v_P v_P^{\mathsf{T}} \succeq 0$ and $\Delta = \det C \geq 0$. □

C. Special cases

Depolarizing. $\lambda_a = \lambda_b = \lambda_c = \eta = 1 - 4p$ gives $X = \sqrt{\eta}\mathbb{1}$ and $Y = (1-\eta^2)/(2\eta)\mathbb{1} = (1-\eta)(N_E + \frac{1}{2})\mathbb{1}$ with $N_E = 1/(2\eta)$. This is the thermal attenuator of Ahmed *et al.*, which they write as $\mathcal{A}_G \circ \mathcal{L}_T$ with $T = 2\eta^2/(1+\eta)$ and $G = (1+\eta)/(2\eta)$ [11].

Phase-covariant channels. If $\lambda_a = \lambda_b = \lambda_\perp$ and $\lambda_c = \lambda_\parallel$ (pole on the axis of phase covariance), the emergent channel is phase insensitive, with

$$\tau = \frac{\lambda_\perp^2}{\lambda_\parallel}, \qquad Y = \frac{1-\lambda_\perp^2}{2\lambda_\parallel}. \tag{8}$$

Every such channel equals $\mathcal{A}_G \circ \mathcal{L}_T$ with $G = Y + (1+\tau)/2$ and $T = \tau/G$.

BB84. With the pole on y: $\lambda_x = \lambda_z = s := 1 - 2q$ and $\lambda_y = s^2$. Then $\tau = 1$, and the emergent channel is the AGN channel with

$$\sigma^2 = Y = \frac{1-s^2}{2s^2} = \frac{1-(1-2q)^2}{2(1-2q)^2}. \tag{9}$$

The map $q \mapsto \sigma^2$ is increasing. Gaussian inputs stop at $\sigma^2 = 1/e$, i.e. $q = 0.120488$; antidegradability starts at $\sigma^2 = 1/2$, i.e. $q = 0.146447$; and $\sigma^2 = 2/5$ corresponds to $q = 0.127322$.

2-Pauli. With the pole on y: $\lambda_x = \lambda_z = 1 - 2p$ and $\lambda_y = 1 - 4p$, so $\tau = (1-2p)^2/(1-4p) > 1$ and $Y = 2p(1-p)/(1-4p)$. This is $\mathcal{A}_G \circ \mathcal{L}_T$ with $T = 1 - 2p$ and $G = (1-2p)/(1-4p)$, a quantum-limited amplifier after pure loss.

Remark 1 (A heuristic mechanism). With the pole on the phase-covariance axis, a unital phase-covariant Pauli channel is the mixture

$$\mathcal{N} = \lambda_\perp\,\mathrm{id} + (\lambda_\perp - \lambda_\parallel)\,F + (1 - 2\lambda_\perp + \lambda_\parallel)\,R \tag{10}$$

whenever the coefficients are nonnegative. Here $R(\rho) = \mathrm{Tr}\,\rho\,\mathbb{1}/2$ and $F(\rho) = \frac{1}{2}(\sigma_a\rho\sigma_a + \sigma_b\rho\sigma_b)$ ("measure σ_c, then flip"). BB84 has coefficients $(s,\, s(1-s),\, (1-s)^2)$ and 2-Pauli has $(1-2p,\, 2p,\, 0)$. Apply F to a random fraction f of the qubits of a symmetric state with few excitations. It removes the excitations it finds, which acts as pure loss with $T = 1 - f$, and it appends a block polarized toward the opposite pole. After recoupling to the total spin, the Holstein–Primakoff mode of that block enters phase-conjugated, which acts as a quantum-limited amplifier with $G = (1-f)/(1-2f)$. For $f = 2p$ this gives the 2-Pauli emergent channel above. It is the analogue of the "loss plus universal cloning" decomposition on which the depolarizing proof of Ref. [11] rests (their Theorem 2), with the cloner replaced by coupling to a flipped block. We do not turn this into a proof here.

D. Numerical validation

Table II compares the exact M-qubit coherent information (Sec. IV, double precision) for thermal (geometric) weights on the Dicke ladder of the BB84 channel with the prediction of the emergent channel. The gap decays as c/M with $c \approx 1.25\,\bar{n}$, and Richardson extrapolation of the $q = 0.1184$ column gives $+0.0080$, against the predicted $+0.0077$. These numbers are floating-point evidence for Proposition 1, not certificates.

III. CEILING OF THE GAUSSIAN ROUTE

Lemma 2. *A phase-insensitive single-mode Gaussian channel with transmissivity or gain $\tau > 0$ and added noise Y (vacuum variance $1/2$) is antidegradable if and only if $Y \geq \tau/2$.*

TABLE II. BB84 channel, thermal input with mean excitation $\bar{n}$ on the Dicke ladder (truncated): exact I_c (bits) of the M-qubit state versus the emergent-channel prediction (double precision; not certified).

M	$q = 0.10$, $\bar{n} = 1$	$q = 0.1184$, $\bar{n} = 4$
8	0.0922	
16	0.1203	
32	0.1427	
64	0.1546	−0.0711
128		−0.0314
256		−0.0117
Richardson ($M \to \infty$)		+0.0080
emergent channel	0.1668	+0.0077

Proof. Lami *et al.* [15] give necessary and sufficient conditions for two-extendibility, i.e. antidegradability, of the three classes, recovering Ref. [14]. In our units: thermal attenuator, $Y = (1-\tau)(N+\frac{1}{2})$, iff $\tau \le (N+\frac{1}{2})/(N+1)$; amplifier, $Y = (\tau-1)(N+\frac{1}{2})$, iff $\tau \ge (N+\frac{1}{2})/N$; additive noise ($\tau = 1$) iff $Y \ge \frac{1}{2}$. Substituting N, each condition is $Y \ge \tau/2$. □

Corollary 1. *The emergent channel (8) of a phase-covariant Pauli channel is antidegradable if and only if $\lambda_\perp \le 1/\sqrt{2}$, independently of $\lambda_\parallel$.*

Proof. $(1-\lambda_\perp^2)/(2\lambda_\parallel) \ge \lambda_\perp^2/(2\lambda_\parallel) \iff \lambda_\perp^2 \le \frac{1}{2}$. □

The value $1/\sqrt{2}$ is the shrinking factor of the optimal phase-covariant $1 \to 2$ qubit cloner (fidelity $\frac{1}{2} + \sqrt{1/8}$) [21]. Codes analysed through the emergent channel, i.e. with a bounded number of excitations near one pole as $M \to \infty$, therefore cannot certify $Q > 0$ beyond the phase-covariant cloning point. For BB84 and 2-Pauli this point is $q, p = (1 - 1/\sqrt{2})/2 = 0.146447$, the familiar 14.6% error rate of the phase-covariant cloning attack on BB84 [21]. For the depolarizing channel it is $3p = \frac{3}{4}(1-1/\sqrt{2}) = 0.219670$, below the true antidegradability point $3p = 1/4$ set by universal cloning. For the AGN channel it is $\sigma^2 = 1/2$. Explicit finite-M states are not bound by this ceiling.

For phase-covariant noise with the pole on the third axis, the ceiling exceeds the hashing point only when the noise along the pole is not too large. Table III shows this for Z-biased noise $(p_X, p_Y, p_Z) = p\,(\frac{1}{2(b+1)}, \frac{1}{2(b+1)}, \frac{b}{b+1})$ with the pole on z: the route is useless for $b \gtrsim 1$ and useful between the 2-Pauli ($b = 0$) and depolarizing ($b = \frac{1}{2}$) channels.

TABLE III. Z-biased Pauli noise with total error p and pole on z: hashing point, the Gaussian-route threshold with thermal inputs at infinite energy (closed form, Holevo–Werner type [12]), and the ceiling of Corollary 1.

bias b	hashing	Gaussian inputs	ceiling
0 (2-Pauli type)	0.22709	0.24149	0.29289
0.25	0.19260	0.20214	0.24408
0.5 (depolarizing)	0.18929	0.18470	0.21967
1	0.19378	0.16856	0.19526
3	0.22221	0.15234	0.16737
10	0.27791	0.14643	0.15342

IV. EXACT FINITE-M COMPUTATION

Generator. With the pole on the phase-covariance axis c (the y axis for BB84 and 2-Pauli, the z axis for the depolarizing channel and the family of Sec. VI A), the channel is unital and phase covariant, and

$$\mathcal{N} = e^{\mathcal{L}}, \quad \mathcal{L} = g\,\mathcal{D}[\sigma_-] + g\,\mathcal{D}[\sigma_+] + \tfrac{\gamma_D}{4}\,\mathcal{D}[\sigma_c], \qquad (11)$$

with $g = -\frac{1}{2}\ln\lambda_\parallel$ and $\gamma_D = -2\ln\lambda_\perp + \ln\lambda_\parallel$, where $\sigma_\pm$ raise and lower along c. For 2-Pauli $\gamma_D < 0$: $\mathcal{L}$ is then not a Lindbladian, but $e^{\mathcal{L}}$ is still the CPTP map $\mathcal{N}$. On M qubits $\mathcal{N}^{\otimes M} = \exp(\sum_i \mathcal{L}_i)$.

Dicke basis and bands. On permutation-invariant operators, $\sum_i \mathcal{L}_i$ acts on the coefficients of $|J,m\rangle\langle J,m'|$ through the closed-form coefficients of the PIQS method [22]. Each J couples only to J and $J \pm 1$. In our installation of `qutip.piqs` [23] these coefficients came out with single-precision accuracy (relative error $\sim 10^{-8}$). We therefore reimplemented them in double precision and, for the certificates, in ball arithmetic. Phase covariance conserves $\delta = m - m'$, so each band δ is an invariant subspace of dimension $\sum_J (2J+1-\delta) \approx M^2/4$, instead of the $\sim M^3/6$ of the full permutation-invariant operator space.

Codes and coherent information. The input is a rank-two state $\rho = (|\phi_0\rangle\langle\phi_0| + |\phi_1\rangle\langle\phi_1|)/Z$ in the $J = M/2$ block, with unnormalized codewords $\phi_i = \sum_{k \le K} c_i(k)|M/2, M/2 - k\rangle$. We use residue codes: $c_i(k) = 0$ unless $k \equiv a_i \pmod\kappa$, with $\kappa = 2$ (parity) or 3 and $(a_0, a_1) = (0, 1)$. Because the channel is linear, the outputs of all $|k\rangle\langle l|$ with $k, l \le K$ are computed once, band by band, and every input is evaluated by contraction. The phase covariance makes B_J block diagonal in $x \bmod \kappa$ (x = excitations from the top of the spin-J block) and $(RB)_J$ block diagonal in $(x - a_i) \bmod \kappa$, and I_c follows from Eq. (4). Nothing is truncated: Sym^M and all output sectors are finite dimensional.

Optimization and validation. Real amplitudes are optimized with L-BFGS using exact gradients obtained from the adjoint map. Thresholds are located by continuation and bisection in p, with warm starts and the criterion $I_c > 10^{-6}$ bits. The band evolution agrees with brute-force 2^M simulation of random rank-two inputs for $M = 3, \ldots, 6$ (BB84, depolarizing and 2-Pauli) to $|\Delta I_c| \le 1.5 \times 10^{-15}$. From a parity-code start at $n = 30$ the optimizer finds $I_c > 0$ at $q = 0.1185$ (BB84) and $p = 0.118$ (2-Pauli), consistent with Ref. [9]. At the

2-Pauli and BB84 $M = 60$ threshold points, an independent evaluation with `qutip.piqs` gives I_c about 1×10^{-7} below ours, a difference explained by the single-precision coefficients.

V. RIGOROUS CERTIFICATION

Every certified point uses an exact rational noise parameter (e.g. $q = 63/500$), hence rational Bloch factors. The stored float64 amplitudes are read as exact dyadic rationals, so each certificate concerns one exactly specified state. Certifier B rejects inputs with weight off the declared residues or with $K > M$; for the inputs of certifiers F and L the exact-zero residue pattern was checked separately. Ball (midpoint–radius) arithmetic is provided by Arb [24] through python-flint 0.9.0 [25].

Entropy enclosure (both certifiers). Let A be a ball matrix that contains a density block (of B_J or $(RB)_J$, in one residue sector). Take a floating-point approximate eigenbasis V, read exactly, and form $A' = V^{\mathsf{T}} A V = \mathrm{diag}(d) + E$ in ball arithmetic. Weyl's inequality gives $|\lambda_i(A') - d_{(i)}| \le \|E\|_F$. Ostrowski's theorem [26] gives $\lambda_i(A') = \theta_i \lambda_i(A)$ with $|\theta_i - 1| \le \delta$, where $\delta \ge \|V^{\mathsf{T}} V - \mathbb{1}\|_F$ and $\delta < \frac{1}{2}$ is checked. Since the true block is positive semidefinite, every eigenvalue is enclosed in an interval in $[0, \infty)$, and $h(t) = -t \ln t$, monotone on $[0, 1/e]$ and on $[1/e, 1]$, gives a rigorous interval for the entropy.

Certifier F (`certify_finite.py`). All generator entries, including the rates $-\frac{1}{2}\ln\lambda$ and square roots, are Arb balls (160 bits). Each band is exponentiated with `arb_mat.exp`, which returns a rigorous enclosure. The blocks are assembled in ball arithmetic and then enclosed as above. The cost is $\sim n^3$ per band of size n: 2.5×10^3 s for BB84 at $M = 60$, practical up to $M \approx 60$.

Certifier B (`certify_big.py`). This certifier integrates the band ODE $v' = Lv$ in floating point with a rigorous a posteriori budget in the 1-norm. All bounds below are per column of the propagated input operators.

1. *Generator.* The entries of L are Arb balls (128 bits), and the float matrix is $L_f = \mathrm{fl}(\mathrm{mid}\, L)$, with $|L_f - L| \le \mathrm{rad} + u|L_f|$ entrywise. This yields rigorous bounds on $\|L - L_f\|_1$, $\|L_f\|_1$ and the logarithmic norm $\mu_1(L_f) = \max_c \big(L_{f,cc} + \sum_{r \ne c} |L_{f,rc}|\big)$ [27], so that $\|e^{tL_f}\|_1 \le e^{t \max(\mu_1, 0)}$.

2. *Integration.* $e^{L_f} = (e^{hL_f})^s$ with $h = 2^{-k}$ and $h\|L_f\|_1 \le \frac{1}{4}$ (cf. Ref. [28]). Each step is a degree-16 Taylor sum. The budget bounds the truncation (series tail, doubled), the rounding of every sparse product, $\gamma_q |L_f||t|$ with q the maximal number of nonzeros per row plus one [29], propagated through the remaining Taylor recursion, and the rounding of the summation. Step errors are propagated to $t = 1$ with $e^{h \max(\mu_1, 0)}$ per step.

3. *Model error.* Duhamel's formula gives $\|e^L - e^{L_f}\|_1 \le e^{\max(\mu_1(L), \mu_1(L_f), 0)} \|L - L_f\|_1$, with $\mu_1(L) \le \mu_1(L_f) + \|L - L_f\|_1$.

4. *Assembly and entropy.* The float blocks σ are exact dyadic matrices. Their distance to the true blocks is at most ε in total absolute entry sum, hence in trace norm. Here ε is the sum of all terms above, weighted by $|c_a(k) c_b(l)|$, plus the assembly rounding, divided by a lower bound on Z. The entropy of σ is enclosed as above. With the negative eigenvalues of σ clipped, the true spectrum is within $\varepsilon' = \varepsilon +$ (negative mass) in ℓ_1, and the continuity bound $|\sum h(\lambda_i) - \sum h(x_i)| \le \varepsilon' \ln(D/\varepsilon')$ for $\varepsilon' \le 1/e$ (cf. Ref. [30]) closes the interval, with D the total number of eigenvalues. A factor of 2 is applied to all budget totals.

For BB84 and the depolarizing channel the band generators have $\mu_1 \le 0$, so the evolution contracts in these coordinates. For 2-Pauli $\gamma_D < 0$, and μ_1 grows roughly linearly with M (≈ 3.2, 6.8, 10.6, 14 at $M = 60, 120, 180, 240$), so float64 rounding can be amplified by up to e^{μ_1}. Certifier B was validated against certifier F on the three $M = 60$ states: the intervals of B contain those of F. At $M = 240$ it took 4.3×10^3 s (depolarizing) and 8.0×10^3 s (BB84) of wall time on a 4-core x86-64 machine.

Extended precision and a self-found bug. With `CERT_LD=1` the integration runs in x87 extended precision ($u = 2^{-64}$). The certifier refuses this mode unless `numpy.longdouble` has a 64-bit mantissa. At $M = 60$ this mode gives $[4.4142155, 4.4142560] \times 10^{-6}$ for the 2-Pauli state, about 10^3 times tighter than float64 and containing the Arb-exponential interval. A self-audit found that the first version of this mode formed the Taylor step scalar h/n as a float64 quotient. That rounding, of relative size up to 2^{-53}, was not covered by the 2^{-64} budget. The scalar is now formed in extended precision, and the fix was made before any extended-precision result entered the manifest. The $M = 60$ cross-check was repeated after the fix, with an interval containing the Arb-exponential one, and the two $M = 180$ runs started with the faulty version were discarded. Float64 mode was not affected, because its $4u$ term covers that rounding.

Certifier L (`certify_limit.py`, bosonic). For channels $\mathcal{A}_G \circ \mathcal{L}_T$ with exact rational T, G, the Fock-basis band matrix elements

$$\mathcal{L}_T(|m\rangle\langle m+d|) = \sum_k \ell_k \, |m-k\rangle\langle m-k+d|,$$
$$\ell_k = \sqrt{\tbinom{m}{k}\tbinom{m+d}{k}}\, T^{m+\frac{d}{2}-k}(1-T)^k,$$
$$\mathcal{A}_G(|j\rangle\langle j+d|) = \sum_k a_k \, |j+k\rangle\langle j+k+d|,$$
$$a_k = \sqrt{\tbinom{j+k}{k}\tbinom{j+d+k}{k}}\,(1-G^{-1})^k\, G^{-(j+\frac{d}{2}+1)},$$

are evaluated in Arb (192 bits) for inputs supported on $n \le N_{\mathrm{in}}$. Output levels above D are collapsed into a flag state $|\perp\rangle$: $\mathcal{T}_D(\omega) = P\omega P + \mathrm{Tr}[(\mathbb{1} - P)\omega]|\perp\rangle\langle\perp|$. $\mathcal{T}_D$ is a channel on the output, so data processing gives $I_c(\rho, \mathcal{T}_D \circ \mathcal{N}) \le I_c(\rho, \mathcal{N})$. Off-diagonal flag terms between

codewords of different residues vanish by phase covariance. The flag entropies are enclosed by monotonicity of h, which stays valid when a flag mass is below the working precision. Restricted to inputs on $\mathrm{span}\{|0\rangle,\ldots,|N_{\mathrm{in}}\rangle\}$, $\mathcal{T}_D\circ\mathcal{N}$ is a channel between finite-dimensional spaces. Hence $Q(\mathcal{N})\geq Q(\mathcal{T}_D\circ\mathcal{N}|_{\leq N_{\mathrm{in}}})\geq I_c(\rho,\mathcal{T}_D\circ\mathcal{N})$ by the finite-dimensional coding theorem [1, 2], with no infinite-dimensional subtleties.

Independent reviews. Two reviews were carried out by separate AI-agent instances started with fresh context; they are not human reviews. The first attacked the three $M=60$ certificates. It re-derived the frame change and Bloch factors, checked $e^{\mathcal{L}}=\mathcal{N}$ including the negative-dephasing case, and compared the generator line by line with `qutip.piqs`. It verified the band evolution at $M=60$ against an exact Schur–Weyl formula for product inputs (agreement $\sim 10^{-14}$), compared its own brute force with both certifiers for $M\leq 10$, checked the exact-zero residue pattern of the stored states, and fuzzed the entropy enclosure (219 cases). Its verdict was that all three certificates are valid. Its minor findings (outward rounding of quoted intervals, a fail-safe NaN case, provenance of logs) were fixed. The second reviewed certifier B line by line against its budget, including the trace-norm/Mirsky/Fannes chain and the multiplicity bookkeeping. It compared its own brute force for 13 random residue codes ($M=4$–10, all channel types) with both certifiers in both precisions, and compared B with F at $M=24$ and 30. It measured the true integration error against Arb references for 2-Pauli bands with μ_1 up to 76: at most 1% of the claimed bound in float64 and 29% in extended precision. It also re-certified the family point $r=1/4$ with F inside the published interval. Its verdict was that float64 mode is sound, including the older code revisions that produced the earlier logs, and that extended-precision mode is sound on this host after the step-scalar fix. Its suggestions were applied: a platform check for extended precision, explicit errors instead of assertions, input validation, and a re-run of the certificates with the final code.

Trust assumptions. The remaining assumptions are: correctness of the certifier code; correctness of python-flint/Arb; IEEE 754 round-to-nearest and SciPy's CSR product semantics (no duplicate entries); x87 extended precision for the one entry computed with `CERT_LD=1` (checked at run time); and the PIQS coefficient formulas [22], checked against brute force for $M\leq 10$ and against an exact Schur–Weyl formula at $M=60$.

VI. CERTIFIED THRESHOLDS

Table IV lists every certificate. Each row is an explicit permutation-symmetric rank-two state on M qubits with a rigorous, outward-rounded interval for $I_c(\rho,\mathcal{N}^{\otimes M})$, hence $Q\geq I_c/M>0$ and $p_{\mathrm{th}}\geq p$. No limit theorem is used. Figure 1 shows the best certified value per M.

Headline numbers. $p_{\mathrm{th}}(\mathrm{BB84})\geq 0.126$ ($M=240$), $p_{\mathrm{th}}(\text{2-Pauli})\geq 0.1255$ ($M=240$), and, for the depolarizing channel, a certified explicit-size value $3p=0.1977936$ ($M=240$). The last exceeds the previous explicit certified 0.194868 ($n=45$) [10] but not the asymptotic 0.2029161 [11]; in the Gaussian limit we certify 0.199546875 (Sec. VI B). The BB84 and depolarizing families are closed under composition, with the less noisy channel as a factor, so $Q>0$ also at every smaller noise. The 2-Pauli family is not (XZ errors appear), so there $p_{\mathrm{th}}\geq 0.1255$ holds in the sup sense only.

Codes. For BB84 and 2-Pauli, mod-3 residue codes beat parity codes at the same M; for the depolarizing channel we used mod-3 codes throughout, as in Ref. [11]. The sequence $M=30\to 60\to 120$ (BB84 $0.1185\to 0.1217\to 0.1239$, 2-Pauli $0.1180\to 0.1211\to 0.1232$, parity codes) still rises with roughly geometric increments. The depolarizing explicit-M values saturate: 0.06522, 0.06547, 0.06569 per Pauli at $M=60$, 90, 120, well below the asymptotic 0.06764. These sequences are bisection values, certified only where listed in Table IV.

A. A family between 2-Pauli and depolarizing

Let $(p_X,p_Y,p_Z)=p\,(1,1,r)/(2+r)$ with total error p and the pole on z. $r=0$ is the 2-Pauli channel with $p=2p_{2\mathrm{P}}$ (up to relabeling), and $r=1$ is the depolarizing channel with $p=3p_{\mathrm{dep}}$. Table V lists the certified points. The gain over hashing is 0.006–0.023 across the family, and the certified thresholds decrease monotonically from 2-Pauli to depolarizing. We found no point-by-point records for the interior of this family in the literature. Bausch and Leditzky [7] cover the Pauli simplex with graph-state codes; we did not compare point by point.

B. Gaussian limit of the depolarizing channel

We continued the $M=240$ mod-3 code in the Gaussian limit (the thermal attenuator with $\nu=1/(2\eta)$) with I_c and its gradient evaluated in 128-bit ball arithmetic (refined eigenbases, so that line searches are not limited by the $\sim 3\times 10^{-14}$ noise floor of double precision for this problem), at $N_{\mathrm{in}}=240$, $D=360$. At $\eta=2351/3200$, i.e. $p=849/12800$ per Pauli ($3p=0.198984375$; $T=5527201/8881600$, $G=5551/4702$), the code has certified $I_c\in[1.82651528,1.85766717]\times 10^{-8}$ bits (certifier L, flag masses $\sim 10^{-56}$); at $\eta=2939/4000$ ($3p=0.1989375$), $[1.07683723,1.07994494]\times 10^{-7}$. An independent re-implementation (pure loss then amplifier, the thermal master equation, and a beam splitter with a Fock-mixed thermal environment; output cutoff 420 photons) gives 1.8455798×10^{-8} and 1.0787368×10^{-7}, with the routes agreeing to 10^{-13}. With Theorem 6 of Ref. [11] (proved for the depolarizing channel), $p_{\mathrm{th}}\geq 0.198984375$ in total error, for sufficiently large M. The branch ends there for

TABLE IV. Certificate manifest. Codes are rank two with residues $\{0, 1\}$ mod κ, and K is the maximal excitation number. Certifier F: Arb exponential per band; B: float integration with a rigorous budget (float64); B-LD: B in x87 extended precision. "sha256" gives the first 16 hex digits of the hash of the state file (Sec. X). Intervals are in units of 10^{-6} bits and rounded outward. For the depolarizing rows p is per Pauli, and the total error is $3p = 0.1956561$, 0.1970625, 0.1977936 at $M = 60$, 120, 240. The family r has $(p_X, p_Y, p_Z) = p\,(1, 1, r)/(2 + r)$, with p the total error.

channel	p (exact)	p	cert.	M	K	κ	I_c (10^{-6} bits)	sha256 (16)
2-Pauli	605469/5000000	0.1210938	F	60	60	2	[4.41423406, 4.41423749]	`1f5e1e9af7f55fea`
2-Pauli	605469/5000000	0.1210938	B	60	60	2	[4.40445984, 4.42401170]	`1f5e1e9af7f55fea`
BB84	304297/2500000	0.1217188	F	60	60	2	[6.34444364, 6.34445247]	`3eec80bc6be5f2eb`
BB84	304297/2500000	0.1217188	B	60	60	2	[6.34280587, 6.34609023]	`3eec80bc6be5f2eb`
depolarizing	652187/10000000	0.0652187	F	60	60	3	[1.54889076, 1.54890269]	`95b7b71c59885ac9`
depolarizing	652187/10000000	0.0652187	B	60	60	3	[1.54770191, 1.55009153]	`95b7b71c59885ac9`
BB84	619531/5000000	0.1239062	B	120	90	2	[1.44239924, 1.44929662]	`6248359d72a58e80`
2-Pauli	308047/2500000	0.1232188	B	120	90	2	[1.73815583, 2.46289307]	`c54db19b8afc4d82`
depolarizing	1051/16000	0.0656875	B	120	90	3	[1.26617376, 1.27106873]	`2067c9f0cd871ed3`
BB84	156211/1250000	0.1249688	B	120	90	3	[3.32197192, 3.32992534]	`e3e7a1aff721ad41`
2-Pauli	154961/1250000	0.1239688	B	120	90	3	[4.05046902, 4.96028384]	`65593ca6babd71c1`
family $r = 1/4$	17827/80000	0.2228375	B	60	60	3	[1.32667340, 1.33009405]	`1546bd790432b60b`
family $r = 1/2$	3359/16000	0.2099375	B	60	60	3	[3.73849269, 3.74092802]	`3a05874737beccad`
family $r = 3/4$	16127/80000	0.2015875	B	60	60	3	[1.63392704, 1.63632317]	`0ea6c8003a9478ba`
depolarizing	41207/625000	0.0659312	B	240	110	3	[1.32046287, 1.33093447]	`9dffeb3cd424e413`
BB84	63/500	0.126	B	240	110	3	[5.54571350, 5.56189889]	`b947cad42e44ebca`
2-Pauli	2003/16000	0.1251875	B-LD	180	90	3	[1.17881571, 1.19653069]	`29f739e0098b53a8`
2-Pauli	251/2000	0.1255	B-LD	240	110	3	[1.88681426, 2.70734623]	`682e0a3d42e431d1`

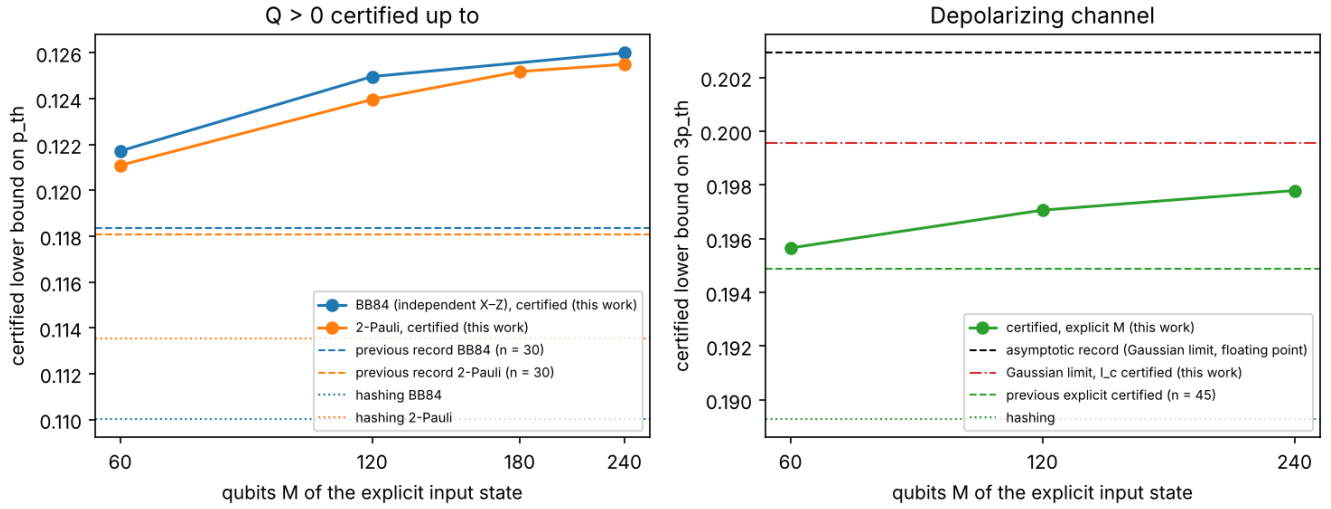


FIG. 1. Certified lower bounds on p_{th} versus the number M of qubits of the explicit input state (best code per M, Table IV). Left: BB84 and 2-Pauli; dashed, the previously reported values at $n = 30$ [9]; dotted, hashing. Right: depolarizing channel in total error $3p$; black dashed, the asymptotic Gaussian-limit value 0.2029161 [11] (floating point); red dash-dotted, our certified Gaussian-limit value 0.199546875 (Sec. VI B); green dashed, the previous certified explicit value 0.194868 ($n = 45$) [10]; dotted, hashing.

a genuine reason: at $\eta = 0.734625$ the optimized value is $I_c = -1.3 \times 10^{-7}$, far above any rounding error, with the mean photon number still near 20.

More photons move the threshold. Stretching that state by a factor 2 in the residue index ($n = 3m + r$, $m \to 2m$; mean photon number 38.5) and optimizing at $\eta = 3673/5000$, where the 20-photon branch is negative, gives $p = 1327/20000$ per Pauli ($3p = 0.19905$; $T = 13490929/21682500$, $G = 8673/7346$) with certified $I_c \in [1.52256332, 1.54665840] \times 10^{-8}$ bits; the independent re-implementation gives 1.5376335×10^{-8}. Continuing this branch, the mean photon number stays at 38.5

TABLE V. Family r: hashing point and certified thresholds (total error p; mod-3 codes; certifier B unless noted).

r	hashing	certified, $M = 60$	certified, larger M
0	0.227092	0.2421876 (parity)	0.2479376 ($M = 120$)
1/4	0.199895	0.2228375	
1/2	0.192599	0.2099375	
3/4	0.189926	0.2015875	
1	0.189290	0.1956561	0.1977936 ($M = 240$)

while I_c keeps growing under further optimization; at $\eta = 7341/10000$, i.e. $p = 2659/40000$ per Pauli ($3p = 0.199425$; $T = 53890281/86705000$, $G = 17341/14682$), the code has certified $I_c \in [4.69714037, 4.72320460] \times 10^{-8}$ bits (independent re-implementation: 4.7133075×10^{-8}; intermediate certified points: $3p = 0.1992375$ and 0.19933125; at $\eta = 0.7335$, 60 L-BFGS steps still leave $I_c = -8.6 \times 10^{-8}$). Stretching once more (by 1.5, mean photon number 57.5) gives a branch that is positive at $\eta = 11743/16000$, i.e. $p = 4257/64000$ per Pauli ($3p = 0.199546875$; $T = 137898049/221944000$, $G = 27743/23486$): certified $I_c \in [2.57112611, 2.97906949] \times 10^{-9}$ bits, independent value 2.8369733×10^{-9}. Hence $p_{\text{th}} \geq 0.199546875$ with Theorem 6 of Ref. [11]. The gain per stretch shrinks ($\eta \approx 0.73467$, 0.7341, 0.73394 for 20, 38.5, 57.5 photons), so stretching alone does not reach $\eta = 0.7294452$: the 56-photon record code must have a different structure. The coherent information shrinks as the photon number grows, which is the trade-off that leads to the 56-photon codes of the floating-point record.

C. Gaussian limit of the 2-Pauli channel

For the emergent 2-Pauli channel at $p = 31/250 = 0.124$ ($T = 94/125$, $G = 94/63$), a mod-4 residue code with codewords on $n \equiv 0, 2 \pmod 4$ ($N_{\text{in}} = 160$, $D = 280$, flag mass 2×10^{-17}) has certified $I_c \in [1.96141999, 1.96142031] \times 10^{-4}$ bits (certifier L). Transferring this to the qubit channel would require extending the convergence theorem of Ref. [11] to the 2-Pauli channel. The explicit $M = 180$ certificate (0.1251875) makes that step unnecessary for the record.

VII. THE ADDITIVE GAUSSIAN NOISE CHANNEL BEYOND $\sigma^2 = 1/e$

By Eq. (9), the Gaussian limit of the BB84 channel is the AGN channel $\mathcal{N}_\sigma$: $V \mapsto V + \sigma^2 \mathbb{1}$, $X = \mathbb{1}$. It agrees with the Gaussian random displacement channel of Ref. [16] ($\hat{q} = (a + a^\dagger)/\sqrt{2}$, covariance shifted by $\sigma^2 \mathbb{1}$). Moreover,

$$\mathcal{N}_\sigma = \mathcal{A}_G \circ \mathcal{L}_T, \qquad G = 1 + \sigma^2, \quad T = 1/G, \tag{12}$$

since both sides are Gaussian with $X = \sqrt{GT}\,\mathbb{1} = \mathbb{1}$ and $Y = \frac{G(1-T)}{2} + \frac{G-1}{2} = G - 1 = \sigma^2$. Since $\mathcal{N}_{\sigma_2} \circ \mathcal{N}_{\sigma_1} = \mathcal{N}_{\sqrt{\sigma_1^2 + \sigma_2^2}}$, data processing makes $Q(\mathcal{N}_\sigma)$ nonincreasing in σ.

Known bounds (Sec. I): $Q(\mathcal{N}_\sigma) > 0$ for $\sigma^2 < 1/e$ (thermal inputs [12], GKP codes [13]) and $Q(\mathcal{N}_\sigma) = 0$ for $\sigma^2 \geq \frac{1}{2}$ [14, 15]. Explicit GKP codes with maximum-likelihood decoding reach $\sigma = 0.598$ [16].

Theorem 1 (computer-assisted)**.** *Let $\rho = (|\phi_0\rangle\langle\phi_0| + |\phi_1\rangle\langle\phi_1|)/Z$, where the real amplitudes of ϕ_0 (supported on $n \equiv 0 \pmod 3$) and ϕ_1 (on $n \equiv 1 \pmod 3$), $n \leq 140$, are the float64 numbers stored in the file* `agn_k3_sigma2_0.4.npy` *(sha256* `7917b7de2ccf2add…`*; Sec. X). At $\sigma^2 = 2/5$,*

$$I_c(\rho, \mathcal{T}_{320} \circ \mathcal{N}_\sigma) \in [3.35715244,\ 3.35738269] \times 10^{-6} \text{ bits}. \tag{13}$$

Consequently $Q(\mathcal{N}_\sigma) \geq 3.357 \times 10^{-6}$ qubits per channel use at $\sigma^2 = 2/5$. Likewise, the mod-3 state stored in `agn_k3_sigma2_0.405.npy` *(sha256* `7e39e6ffc2144ccc…`*, $n \leq 300$) satisfies, at $\sigma^2 = 81/200$,*

$$I_c(\rho', \mathcal{T}_{600} \circ \mathcal{N}_\sigma) \in [6.33597116,\ 6.40767279] \times 10^{-8} \text{ bits}, \tag{14}$$

so $Q(\mathcal{N}_\sigma) > 0$ for every $\sigma^2 \leq 81/200$. Finally, the rank-two input ρ'' without residue structure stored in `agn_gen_sigma2_0.407.npy` *(sha256* `cb6c059e736810ac…`*, $n \leq 200$) satisfies, at $\sigma^2 = 407/1000$,*

$$I_c(\rho'', \mathcal{T}_{400} \circ \mathcal{N}_\sigma) \in [5.43799903,\ 5.50010769] \times 10^{-8} \text{ bits}, \tag{15}$$

and its continuation ρ''' (`agn_gen_sigma2_0.4075.npy`*, sha256* `756c07a446417776b…`*, $n \leq 300$) satisfies, at $\sigma^2 = 163/400$,*

$$I_c(\rho''', \mathcal{T}_{600} \circ \mathcal{N}_\sigma) \in [5.97733793,\ 6.88032591] \times 10^{-9} \text{ bits}, \tag{16}$$

so $Q(\mathcal{N}_\sigma) > 0$ for every $\sigma^2 \leq 163/400$, i.e. $\sigma \leq 0.6384$.

Certificate. Certifier L (Sec. V) with the exact decomposition (12), $G = 7/5$, $T = 5/7$, $N_{\text{in}} = 140$ and $D = 320$. The flag masses are 3.4×10^{-47} and 3.9×10^{-47}, and the widths of the entropy enclosures are below 6×10^{-11} nats per residue block. The consequences follow from data processing (Sec. V) and the monotonicity in σ. □

The code. The two codewords carry weights 0.446 and 0.554, and the state has mean photon number 15.4. It was obtained by transferring the $M = 240$ BB84 mod-3 code (Table IV, excitations $k \mapsto$ Fock n) to the emergent channel and re-optimizing it by continuation in q. That optimization ran in floating point up to $q = 0.1279375$ ($\sigma^2 \approx 0.40298$), and the resulting state was certified, without further changes, at the slightly smaller noise $\sigma^2 = 2/5$. The truncation is generous: the same certificate also passes with $D = 380$. A re-run of the certifier with single-threaded BLAS gave the nested interval $[3.35715258, 3.35738262] \times 10^{-6}$. The enclosure endpoints depend on the floating-point approximate eigenbasis, and every such interval is rigorous.

Independent check. A separate AI-agent instance, without importing any repository code, implemented the channel as the diffusion semigroup $\exp[\sigma^2(\mathcal{D}[a]+\mathcal{D}[a^\dagger])]$. It checked that this adds exactly σ^2 to both quadrature variances and maps the vacuum to a thermal state with $\bar{n}=\sigma^2$. It cross-checked the Fock matrix elements four ways to 10^{-15} (high-precision quadrature of the displacement average, the Lindblad propagator, $\mathcal{A}_G\circ\mathcal{L}_T$ and $\mathcal{L}_{T'}\circ\mathcal{A}_{G'}$), and obtained $I_c = 3.3572969\times10^{-6}$ bits for the stored state at $\sigma^2=2/5$, stable for cutoffs 320, 400 and 520 and inside the certified interval. Its values also agree with ours at $\sigma^2 = 0.38$, 0.403, 0.404 and 0.405; at 0.405 the coherent information of this state is negative. For the second state it obtained $I_c = 6.38363\times10^{-8}$ bits at $\sigma^2=81/200$ (cutoffs 600, 700 and 800 agree to 3×10^{-13}), inside the certified interval, and reproduced the certificate exactly; the coherent information of that state changes sign at $\sigma^2\approx0.40509$. This is a floating-point re-computation, not a second rigorous certificate.

How far it goes. Re-optimizing the mod-3 family with a larger cutoff ($N_{\rm in}=300$, $D=600$) gave the second state of Theorem 1 (mean photon number 15.4), certified at $\sigma^2=81/200$ with flag masses below 6×10^{-56}; in floating point its coherent information is 1.08×10^{-6} at $\sigma^2=0.4035$ and 6.4×10^{-8} at 0.405. Other residue families with the same cutoff stop earlier in floating point (coherent information above 10^{-6} required): mod 4 $\{0,2\}$ (about 72 photons) at 0.40175, parity and mod 5 $\{0,1\}$ at 0.39975, mod 5 $\{0,2\}$ at 0.39575, mod 4 $\{0,1\}$ at 0.39375 and mod 6 $\{0,3\}$ at 0.39175. Rank-three mod-3 inputs do not help: the optimum stays of rank two.

Breaking the symmetry. Optimizing over all rank-two inputs ($N_{\rm in}=200$, $D=400$), starting from the mod-3 state plus a small random perturbation, gives states that remain about 99.9% on the residues but carry about 0.1% of weight off them in each codeword (overlap $\langle\phi_0|\phi_1\rangle\approx-7\times10^{-4}$). This small breaking of the $\mathbb{Z}_3$ symmetry multiplies the coherent information: 4.37×10^{-7} instead of 6.4×10^{-8} at $\sigma^2=0.405$ (certified: $[4.36362113, 4.37080935]\times10^{-7}$ bits), and continuation in σ^2 keeps it positive up to 0.407 (third certificate) and, with $N_{\rm in}=300$ and $D=600$, to 0.4075 (fourth certificate); it is negative at 0.408. Starting instead from the high-energy mod-4 $\{0,2\}$ state (about 77 photons) with broken symmetry stops at 0.406. Without residue structure the output of $\mathrm{R}\otimes\mathrm{B}$ is certified as a single 802×802 block, and the flag block of $\mathrm{R}\otimes\mathrm{B}$ is no longer diagonal: its entropy is enclosed by $[h(t), h(t)+t\ln2]$, with t the total flag mass ($\approx 7\times10^{-44}$), which holds for any 2×2 positive matrix of trace t. So the residue codes are not local maxima in the full input space, and the true threshold of single-mode rank-two inputs is somewhat above what residue families suggest. The optimal codewords are not three-component cat codes, which give $I_c<10^{-12}$ for every $\sigma^2\in[0.36,0.39]$ after optimizing the amplitude and weights; they have a positive main lobe at $n\approx3$–18 and a tail of alternating sign, with photon-number variance about 2.7 times the Poisson value. A three-parameter physical family explains most of the gain: *squeezed cats*, the $n\equiv0$ and $n\equiv1 \pmod 3$ parts of a displaced phase-squeezed state $D(\alpha)S(r)|0\rangle$ with weights $w, 1-w$. With $\alpha\approx5.71$, $r\approx-1.03$, $w\approx0.452$ it is positive up to $\sigma^2=0.392$; at $\sigma^2=49/125$ (file `agn_sqcat_sigma2_0.392.npy`, sha256 `eb13a99ef8255ad0`…) we certify $I_c\in[1.73865751, 1.73880842]\times10^{-6}$ bits. So about 60% of the improvement over $1/e$ comes from a state of a kind already prepared in optical experiments. The independent re-implementation, with full matrices and two different purifications, gives 4.3682405×10^{-7} and 5.4779650×10^{-8} bits for the two symmetry-free states, and 6.58088×10^{-9} bits (three routes) for the fourth, all inside the certified intervals. The interval $0.4075<\sigma^2<1/2$ remains open.

Consequences. At $\sigma^2=2/5$ the coherent information optimized over Gaussian inputs is zero, while $Q(\mathcal{N}_\sigma)\geq3.357\times10^{-6}$. Hence the Gaussian-optimized one-shot coherent information is not the quantum capacity of the additive Gaussian noise channel, which settles in the negative the speculation of Harrington and Preskill quoted in Sec. I, and the one-shot coherent information itself is strictly larger than its Gaussian restriction there. To our knowledge this is the first rigorous proof of $Q(\mathcal{N}_\sigma)>0$ for any $\sigma>1/\sqrt{e}$; the simulation-based threshold $\sigma\approx0.67$ of Ref. [17], if confirmed, would go further.

Relation to the thermal attenuator. Write $\mathcal{E}_{\eta,\nu}$ for the thermal attenuator with transmissivity η and ν thermal photons, $Y=(1-\eta)(\nu+\frac12)$. Two exact Gaussian identities connect it to $\mathcal{N}_\sigma$:

$$\begin{aligned}\mathcal{E}_{\eta,\nu}&=\mathcal{L}_\eta\circ\mathcal{N}_{\sigma'},\quad &\sigma'^2&=(1-\eta)\nu/\eta,\\ \mathcal{A}_{1/\eta}\circ\mathcal{E}_{\eta,\nu}&=\mathcal{N}_{\sigma''},\quad &\sigma''^2&=(1-\eta)(\nu+1)/\eta.\end{aligned}\tag{17}$$

The first identity shows that the result of Mele *et al.* [18] implies positivity for $\mathcal{N}_{\sigma'}$ only. At their point $\nu=1$, $\eta=0.7841$ this is $\sigma'^2=0.275<1/e$, so it does not imply Theorem 1. Conversely, the second identity and Theorem 1 give $Q(\mathcal{E}_{\eta,\nu})>0$ whenever $(1-\eta)(\nu+1)/\eta\leq0.4075$. For $\nu>4.23$ this region extends below the Holevo–Werner thermal-state threshold $\log_2\frac{\eta}{1-\eta}=g(\nu)$, with $g(\nu)=(\nu+1)\log_2(\nu+1)-\nu\log_2\nu$; for example, at $\nu=10$ it covers $\eta\in[0.96428, 0.96614)$. We have not compared this with the non-Gaussian results of Ref. [18] away from $\nu=1$. The same argument applies to the thermal amplifier $\mathcal{A}_{\tau,\nu}$ ($X=\sqrt{\tau}\,\mathbb{1}$, $Y=(\tau-1)(\nu+\frac12)\mathbb{1}$, $\tau>1$): $\mathcal{L}_{1/\tau}\circ\mathcal{A}_{\tau,\nu}=\mathcal{N}_{\sigma'''}$ with $\sigma'''^2=(\tau-1)(\nu+1)/\tau$, so $Q(\mathcal{A}_{\tau,\nu})>0$ whenever $(\tau-1)(\nu+1)/\tau\leq0.4075$. In general, a phase-insensitive channel with transmissivity or gain τ and added noise Y has $Q>0$ whenever $(Y+|1-\tau|/2)/\tau\leq0.4075$. For both families the new region lies beyond the Gaussian-input threshold exactly when $2^{g(\nu)}>(\nu+1)/0.4075$, i.e. for $\nu>4.23$; at $\nu=10$ the amplifier window is $\tau\in(1.03632, 1.03847]$.

A thermal amplifier at low noise. The emergent channel of the 2-Pauli family, $\mathcal{A}_G\circ\mathcal{L}_T$ with $T=1-2p$ and $G=(1-2p)/(1-4p)$, is a thermal amplifier, and the corollary above does not reach

the relevant parameters. The mod-3 code transferred from $M = 240$ and continued in the Gaussian limit to $p = 0.12725$ (file `amp_2p3_p0.1275.npy`, sha256 `57ba63a1717f0653`…) gives, at $p = 51/400$, i.e. $T = 149/200$ and $G = 149/98$,

$$I_c \in [9.10836870,\ 9.11017611] \times 10^{-7} \text{ bits.} \tag{18}$$

This channel is the thermal amplifier with gain $\tau = 22201/19600 \approx 1.1327$ and $\nu = 149/51 \approx 2.922$ thermal photons. At this gain thermal inputs give positive coherent information only for $\nu < 2.6533$ ($\log_2 \frac{\tau}{\tau-1} - g(\nu) = -0.1187$ at $\nu = 149/51$), and the channel is antidegradable only for $\nu \geq 3.768$ ($Y \geq \tau/2$). Its additive-noise equivalent has $\sigma^2 = 0.459 > 0.4075$, so this is not a consequence of Theorem 1. We know of no earlier non-Gaussian positivity result for thermal amplifiers. The same independent re-implementation (three routes: the Lindblad semigroup with gain and loss rates, $\mathcal{L}_{T'} \circ \mathcal{A}_{G'}$, and $\mathcal{A}_G \circ \mathcal{L}_T$) gives $I_c = 9.109425 \times 10^{-7}$ bits, inside the interval, and a sweep of thermal and squeezed-thermal inputs in 60-digit arithmetic found no Gaussian input with positive coherent information. Breaking the residue symmetry, as for the AGN channel, a rank-two input stays positive up to $p = 0.128$ and is certified at $p = 16/125$ ($T = 93/125$, $G = 93/61$, i.e. $\tau = 8649/7625 \approx 1.1343$ and $\nu = 93/32 \approx 2.906$): $I_c \in [1.43395294, 1.43876228] \times 10^{-7}$ bits, where thermal inputs need $\nu < 2.6206$ and antidegradability starts at $\nu = 3.7231$. The independent re-implementation gives 1.4369750×10^{-7} bits for this state, inside the interval.

The thermal attenuator with one thermal photon. For $\mathcal{E}_{\eta,\nu}$ with $\nu = 1$, Mele *et al.* [18] certified $Q > 0$ down to $\eta = 0.7841$ (thermal inputs stop at $\eta = 0.8$, antidegradability starts at $\eta = 0.75$; their uncertified numerics suggest $\eta \approx 0.7818$). Our optimizer and certifier apply directly, with $\mathcal{E}_{\eta,\nu} = \mathcal{A}_G \circ \mathcal{L}_T$, $G = (1-\eta)(\nu + \frac{1}{2}) + \frac{1+\eta}{2}$, $T = \eta/G$. Among residue families at low energy, mod 3 $\{0, 1\}$ is best (about 17 photons; parity stops at $\eta = 0.78525$, mod 4 $\{0, 1\}$ at 0.78375). At $\eta = 3133/4000 = 0.78325$ (file `th1_k3_p0.21675.npy`, sha256 `eff871b4adcf91ce`…, $N_{\text{in}} = 80$, $D = 240$),

$$I_c \in [8.27408111,\ 8.27425105] \times 10^{-6} \text{ bits.} \tag{19}$$

Since $\mathcal{E}_{\eta,\nu} = \mathcal{E}_{\eta/\eta',\nu} \circ \mathcal{E}_{\eta',\nu}$, Q is non-decreasing in η, so $Q(\mathcal{E}_{\eta,1}) > 0$ for all $\eta \geq 0.78325$, improving the certified value 0.7841. The independent re-implementation, starting from the beam-splitter definition with a thermal environment of mean photon number ν (as in Ref. [18]) and using four routes, gives $I_c = 8.2742014 \times 10^{-6}$ bits, inside the interval; thermal and squeezed-thermal inputs are all negative at this point. Breaking the residue symmetry goes further: a rank-two input stays positive to $\eta = 0.78$ (file `th1_gen_eta0.78.npy`, sha256 `6c91dfe91ec43e9f`…, $N_{\text{in}} = 120$, $D = 300$), where $I_c \in [2.12631890, 2.16095917] \times 10^{-8}$ bits; the independent re-implementation gives 2.150005×10^{-8}. Hence $Q(\mathcal{E}_{\eta,1}) > 0$ for all $\eta \geq 0.78$, below both the certified 0.7841 and the uncertified numerical 0.7818 of Ref. [18]; the window $0.75 < \eta < 0.78$ remains open.

All temperatures. Ref. [18] treats $\nu = 1$ only. The same codes certify $Q > 0$ inside the open window $(\eta_{\text{AD}}, \eta_{\text{G}})$, with $\eta_{\text{AD}} = (\nu + \frac{1}{2})/(\nu + 1)$ and $\eta_{\text{G}} = 2^{g(\nu)}/(1+2^{g(\nu)})$, at every ν we tried (Table VI and Fig. 2); for $\nu \gtrsim 4.2$ the AGN corollary also applies but the direct certificates go further. Since Q is monotone in η at fixed ν, each row certifies $Q > 0$ for all larger η. The independent re-implementation (three routes per point) gives values inside every interval of Table VI, and it regenerated the squeezed-cat state from (α, r, w) to 4×10^{-16}.

VIII. WHAT DID NOT WORK

The depolarizing record. We did not beat $3p = 0.2029161$; our certified Gaussian-limit value is 0.199546875 (Sec. VI B), and the symmetry breaking that helps for the AGN channel did not help here. Full L-BFGS over residue codes ($\kappa = 2, \ldots, 6$, ranks 2–3) for the emergent thermal attenuator, in float64, reaches $I_c > 0$ only down to $\eta \approx 0.737$ ($3p \approx 0.197$). The record point $\eta = 0.7294452$ needs codes with mean photon number ≈ 56 [11]. In that regime the three phase components are almost orthogonal both for Bob and for the environment, and $I_c \approx 0$ with curvature ≈ -0.16 and gradient $\sim 10^{-6}$. Only $\sim 10^{-11}$ of coherent information is available there, consistent with the 10^{-12} reported in Ref. [11]. In 128-bit arithmetic, mod-3 codes with Gaussian envelopes are negative at $\eta = 0.7294452$ for every mean photon number (40–70), width (2–6) and weight (0.30–0.62) we scanned, with $|I_c|$ between 10^{-5} and 10^{-28} (one is enclosed rigorously as $-4.66625545720 \times 10^{-12}$), and they stay negative up to $\eta = 0.75$; squeezed-cat codes behave the same way. The record code therefore has structure beyond a Gaussian envelope that we have not found, and without it we can neither confirm nor refute the floating-point value. Large-amplitude coherent-state cat codes do worse: the quantum Chernoff overlap exponents of Bob and of the environment cross at $\eta = 0.7757$. With explicit M, the thresholds saturate (Sec. VI): reaching 56 excitations needs M far beyond the 240 used here.

Strongly biased noise. By Table III, the Gaussian route is useless for dephasing-biased noise with $b \gtrsim 1$, where the ceiling is below the hashing point.

Float64 certification for 2-Pauli at large M. Because μ_1 grows with M, the float64 budget for 2-Pauli loses up to e^{μ_1}, and the 2-Pauli entries at $M \geq 180$ require extended precision.

IX. OPEN PROBLEMS

- Prove the Gaussian limit (the analogue of Theorem 6 of Ref. [11]) for phase-covariant Pauli channels, for instance along Remark 1, and for general

TABLE VI. Thermal attenuator with ν thermal photons: certified $Q > 0$ at transmissivity η, with the fraction of the open window $(\eta_{\mathrm{AD}}, \eta_{\mathrm{G}})$ it closes. Inputs are rank two, symmetry-free ($k = 1$, $N_{\mathrm{in}} = 160$, $D = 360$) except $\nu = 5$ (mod 3, $N_{\mathrm{in}} = 260$, $D = 520$) and $\nu = 1$ ($N_{\mathrm{in}} = 120$, $D = 300$).

ν	η_{AD}	η_{G}	certified η	I_c (bits)
1/4	0.6	0.65147	313/500 (50%)	$[2.48378509, 2.48746745] \times 10^{-7}$
1/2	0.66667	0.72207	1393/2000 (46%)	$[6.31451396, 6.36306427] \times 10^{-8}$
1	0.75	0.8	39/50 (40%)	$[2.12631890, 2.16095917] \times 10^{-8}$
2	0.83333	0.87097	8587/10000 (33%)	$[1.14709176, 1.14892934] \times 10^{-7}$
3	0.875	0.90459	1791/2000 (31%)	$[8.31897855, 8.33853440] \times 10^{-8}$
5	0.91667	0.93723	7459/8000 (24%)	$[1.17390097, 1.17400578] \times 10^{-6}$

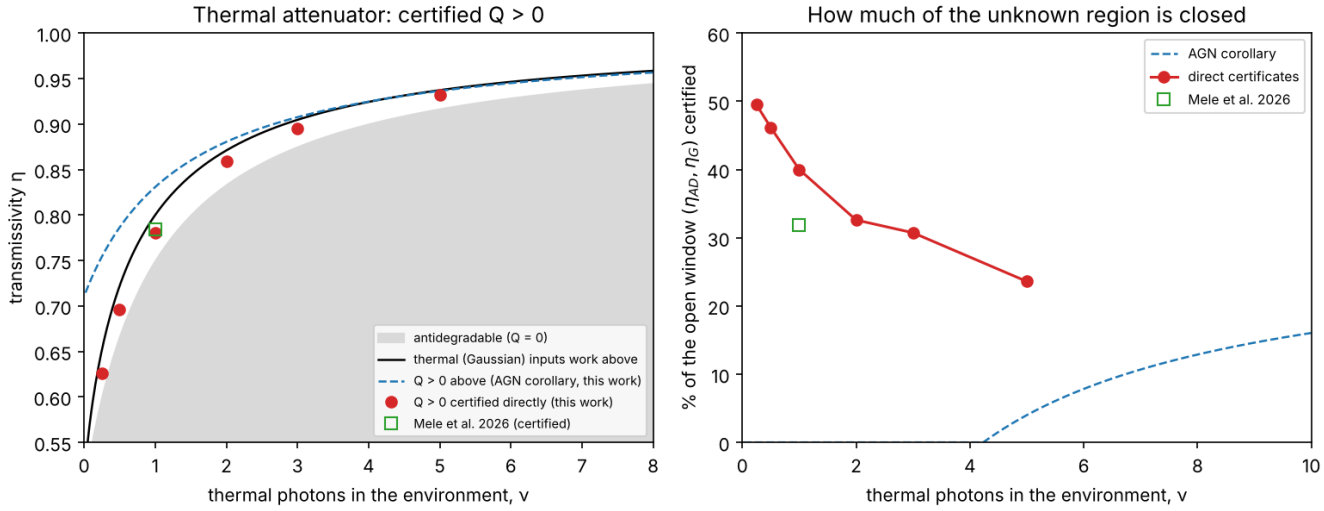


FIG. 2. Thermal attenuator. Left: certified $Q > 0$ above the red points (direct certificates) and above the dashed curve (corollary of Theorem 1); thermal inputs work above the solid curve; gray: antidegradable. Right: fraction of the open window closed at each ν; the square is the certified point of Ref. [18].

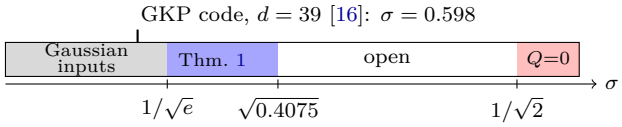


FIG. 3. Positivity of $Q(\mathcal{N}_\sigma)$ along σ. Gray: thermal inputs [12]; blue: this work (Theorem 1); white: open; red: antidegradable. The bar marks the largest σ with a nonzero rate for an explicit GKP code reported in Ref. [16].

Pauli channels with Eq. (6).

- Find the optimal Gaussian-limit codes and turn them into finite-M thresholds. The certified Gaussian-limit values correspond to $q \approx 0.1289$ for BB84 ($\sigma^2 = 0.4075$) and $p = 0.128$ for the 2-Pauli channel (the thermal-amplifier certificate); a convergence theorem for these channels would make them thresholds of the qubit channels. Multi-mode (degenerate) codes are unexplored here. The ceiling is 0.146447.
- Close the AGN window $0.4075 < \sigma^2 < 1/2$: is $Q(\mathcal{N}_\sigma) > 0$ for every $\sigma^2 < 1/2$?
- Push explicit M further: thresholds still rise with M, and certifier B scales like the float computation itself.
- Depolarizing: an explicit-M certificate approaching 0.2029 needs states with ~ 56 excitations, i.e. M in the thousands, or a quantitative convergence theorem.
- An independent, dependency-free verifier in exact integer arithmetic, in the spirit of Ref. [10], and human verification of the present certificates.

X. REPRODUCIBILITY

Code, input states and logs are in the accompanying repository [31], in `pauli-emergence/` and `depolarizing-threshold/`. Requirements are Python

3 with `numpy`, `scipy`, `python-flint` (0.9.0 was used) and `qutip` (only for cross-checks). The runs used Python 3.13, NumPy 2.5.3 and SciPy 1.18.1 on x86-64. The script `recertify_all.sh` runs certifier B with `OMP_NUM_THREADS=1` and bands in parallel (`CERT_PROCS=4`). The arguments are channel, exact p, state file, M, κ and residues (B, F), or channel, parameter, state file, residues, κ, $N_{\rm in}$ and D (L):

```
cd pauli-emergence
python3 certify_finite.py bb84 304297/2500000 \
    runs/cert_bb_M60_thr.npy 60 2 0,1
python3 certify_big.py bb84 63/500 \
    runs/state_bb3_M240_q0.126.npy 240 3 0,1
python3 certify_big.py depol 41207/625000 \
    runs/state_dep_M240_thr.npy 240 3 0,1
CERT_LD=1 CERT_PROCS=4 python3 certify_big.py \
    2pauli 2003/16000 \
    runs/state_2p3_M180_thr.npy 180 3 0,1
python3 certify_limit.py agn 2/5 \
    runs/agn_k3_sigma2_0.4.npy 0,1 3 140 320
python3 certify_limit.py agn 81/200 \
    runs/agn_k3_sigma2_0.405.npy 0,1 3 300 600
python3 certify_limit.py agn 407/1000 \
    runs/agn_gen_sigma2_0.407.npy 0,0 1 200 400
python3 certify_limit.py agn 163/400 \
    runs/agn_gen_sigma2_0.4075.npy 0,0 1 300 600
python3 certify_limit.py 2pauli 31/250 \
    runs/2p_k4_p0.124.npy 0,2 4 160 280
python3 certify_limit.py 2pauli 51/400 \
    runs/amp_2p3_p0.1275.npy 0,1 3 140 320
python3 certify_limit.py 2pauli 16/125 \
    runs/amp_gen_p0.128.npy 0,0 1 200 400
python3 certify_limit.py th1 867/4000 \
    runs/th1_k3_p0.21675.npy 0,1 3 80 240
python3 certify_limit.py th1 11/50 \
    runs/th1_gen_eta0.78.npy 0,0 1 120 300
python3 certify_limit.py agn 49/125 \
    runs/agn_sqcat_sigma2_0.392.npy 0,1 3 140 320
# nu = 1/4, 1/2, 2, 3, 5: see CERTIFICATES.md
```

The full list of commands, state files and hashes is in `CERTIFICATES.md`, and `recertify_all.sh` re-runs all certifier-B entries. State files for the rows of Table IV, in order: `cert_2p_M60_p0.1210938.npy`, `cert_bb_M60_thr.npy`, `cert_dep_M60_thr.npy`, `state_bb_M120_thr.npy`, `state_2p_M120_thr.npy`, `state_dep_M120_thr.npy`, `state_bb3_M120_thr.npy`, `state_2p3_M120_thr.npy`, `state_pr0.25_k3_M60_thr.npy`, `state_pr0.5_k3b_M60_thr.npy`, `state_pr0.75_k3_M60_thr.npy`, `state_dep_M240_thr.npy`, `state_bb3_M240_q0.126.npy`, `state_2p3_M180_thr.npy` (all under `runs/`). Each log in `runs/certB_*.log` records the command line, the input sha256 and the python-flint version. The AGN log (`runs/certL_agn_k3_0.4.log`) does not record the command line; the parameters above were confirmed by re-running it.

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